Assignment 3
Games for Exploring Human Cognition

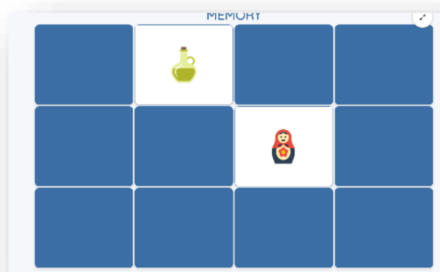
GOALS

This third assignment is meant to give you hands on experience toward the development of a small memory game, continuing in the User Centered Design approach, and applying attention and perception principles, in particular the Gestalt Principles, within your design.

By "memory game" I mean a small game that would make a user practice their memory ability. Here are 3 examples of sites providing memory games. You will notice that their UI is really (really!) not great, and I'm sure you can do better.

Example (1) is a site where a user must find matching cards in a grid.

https://www.helpfulgames.com/subjects/brain-training/memory.html

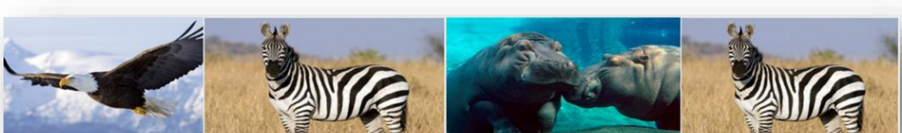


Example (2) is a site for practicing sequence memorization, a *Simon says* game:

https://www.mathsisfun.com/games/simon-says.html).

Example (3), Memory Health Check, is to practice finding pairs in sequences.

https://www.memorylosstest.com/free-short-term-memory-tests-online/



By carrying out this assignment, you will:

- Continue exploring prototyping approaches, in particular the development of mockups with the help of tools, such as Figma.
- Continue in the User Centered Design approach
- Continue exploring visual design, now incorporating knowledge about human cognition (perception, attention, memory)
- Pay attention to Gestalt Laws in Design
- Continue exploring design patterns for Organization, Input and Navigation.
- Learn about game design
- Continue expanding your JavaScript development skills

SCHEDULE / DEADLINES

Here are the important calendar events for this assignment.

- **Mondays, June 8th and June 15th tutorial**

Make sure to attend both June 8th and 15th tutorials to better understand the requirements for this assignment and explore appropriate technologies to help you create your memory game.

- **Forum monitoring on Brightspace between Tuesday June 9th and Friday June 19th**

If something is not clear, if you need help about the assignment, then please ask questions on Brightspace forum. Make sure you get help. Any question is fine, as long as it is asked in a polite and respectful manner. TAs will be monitoring the forum regularly and answer your questions. Not only will their answers help you, they will help other students in the class.

- **Emails to TAs between Tuesday June 9th and Friday June 19th (if needed)**

ONLY if the forum help is not sufficient, you can request a one-on-one help session from the TA assigned to your lab session (make sure you contact the TA for YOUR session – Z01 to Z04).

- **Report submission: Sunday, June 21, midnight**

You MUST submit your report, which will follow the requirements and contain a link to your portfolio by Sunday, June 21, midnight. *No extension will be given. The only exceptions will be for justified medical reasons with medical certificates. The penalty is 20% per day. There is a grace period till 1am… but then it's -20%!*

- **Group review registration: Sunday, June 21, midnight**

Make sure you put your name on the group review schedule. You MUST participate in the group review to practice your analysis skills and get feedback on your design.

- **Group review participation between June 25th and June 30th**

Your scheduled group review session will be within 10 days of your submission. Not registering or missing the group review is 20% off your assignment.

ONLINE TUTORIALS / TECHNOLOGIES

Resources for mockup development and JavaScript development have already been provided in the requirement documents for Assignment 1 and 2.

During the tutorial, the TA will provide examples for game development.

REQUIREMENTS

1. **Explore games and decide on your game**

Spend a bit of time exploring memory games online. Find at least two resources that are inspiring you to develop your game. Decide on the game that you wish to develop. Give a name to your game.

Requirement for your game:

- It must test human memory in some way
- It must contain levels, meaning that the same user would have some option of a level (e.g. beginner, advanced)
- It must contain some possible configuration (e.g. choice of animal or nature images in a pairing game)
- It must be the result of your own understanding and effort. You may use online materials for guidance, but you must not copy from them or from tutorial examples.

2. **Define two personas**

Develop *two personas* as target users of the game. For EACH persona:

- Provide a name to your persona
- Provide 3 intrinsic characteristics to your persona
- Provide the relation to technology and relation to the domain of your persona
- Provide 1 goal to your persona

The chosen goals might vary slightly depending on your type of game.

3. Create 2 different storyboards

Your storyboards will be series of mockups showing how your personas will be playing your game.

Each storyboard should contain at least three mockups showing 3 steps:

- Mockup with selection for level and options
- Mockup with the game being played with feedback (here there might be multiple mockups to show different game situations)
- Mockup for the end of the game (what is displayed to the user)

To practice generating alternatives, you have to create visually and interactively different mockups for your two storyboards. You are required to explore two different color themes, different layouts, different ways of displaying the information. You also are required to explore different ways of interacting with the layout to play.

4. Create one high-fidelity interactive prototype

Using a *JavaScript framework (e.g., React)*, create a single high-fidelity interactive prototype that will combine the goals of your two personas and reflect your visual design choices.

The final design choice can be a combination of the choices made in the two storyboards or could be based on a single storyboard. Make sure you explain the relationship between your storyboards and your final prototype in your report.

The high-fidelity prototype should meet the following technical requirements:

a. Developed using a JavaScript framework (e.g., React)
b. Make use of adequate organization, input and navigation patterns (e.g. patterns facilitating the set up and playing of the game for your users)
c. Be interactive (allow a user to actually play the game)
d. Be accessible through your portfolio (from assignment 1)

Your high-fidelity prototype should be guided by visual design principles:

a. Use a color theme and be consistent
b. Use typography to attract attention
c. Think of the impact of the layout and the use of negative space
d. Use well the principles of contrast, scale, balance, visual hierarchy
e. Use well the Gestalt Principles for Perception (e.g. similarity, continuity, closure, figure and ground, etc).

5. **Link your portfolio to your high-fidelity prototype**

You created your portfolio during Assignment 1. You must now link your portfolio to the high-fidelity prototype of the memory game (as you did for the service site of Assignment 2). You can also (if you wish) link your portfolio to your two storyboards to demonstrate your working method and contribute to the expertise (*how you work*) section of your portfolio.

Your portfolio and memory game should be accessible to the public (or, if you prefer, password protected with the password provided in your report).

The TA must be able to test your UI without having to communicate with you. This is ESSENTIAL.

6. **Write a short report describing your design process and choices:**

Your report should include:

1. Designer: Name, Student Number
2. Game:
    a. What is the name and type of game that you decided to develop
    b. Links to the inspiration sites with a description of how they were used for inspiration.
3. Storyboard with mockups:
    a. Definition of your 2 personas
    b. Your 2 storyboards, one for each persona, and both visually different
4. High-fidelity prototype
    a. An explanation of the visual design choices you made. Make sure to discuss the use of color theme, typography, layout, use of negative space, and application of Gestalt principles.
    b. Link to your portfolio which will contain a link to the memory game
5. Code: A link to the group's GitHub
6. Generative AI Acknowledgement: Many of you will use generative AI to help you create your (a) mockups, (b) high-fidelity prototypes and (c) your report. Describe the role of generative AI for each of those, providing the GenAI tools used and your interactions with them.

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EVALUATION

This assignment is worth 8% of your semester and is evaluated on 100 points.

- Your report's content and clarity (10 points)
  - Report easy to read and answers the required questions.
  - Report contains all the required elements with answers to better understand your design choices and the help provided by GenAI.
  - Different sections of the report are well identified
  - Information is provided in a concise manner, avoiding long paragraphs.
- Your two personas + storyboards (10 points)
- Your hosted web site, accessible, made according to the requirements provided (60 points)
  - UI should reflect the choices made visual design
  - UI should be an extension/combination of storyboards
- Your active participation in the group review (20 points)
  - Ability to analyze other student's design according to the principles explored in the lecture
  - Ability to answer questions from TA and other students

SUBMISSION CONTENT / PLATFORM

Each student must submit their report, PDF version (see all its requirements above) on Brightspace. The link can be found in the Assignment section, for Assignment 3.

ALSO, very important, **MAKE SURE you register on the signup sheet for your group review**. You will be informed when the signup sheet is available.

QUESTIONS

- See the section Schedule/Deadline of the current document. It indicates tutorials and forum for help and questions.
- Please DO NOT send the professor any emails. The forum help should be enough to obtain answers to your questions. Do not be shy to ask questions on the forum. Any answers provided there by the TA will help many other students.
- If the forum help is not sufficient, email the TA assigned to your lab session.